

Proposed Solution

Phase: Project Design

Project: ShopEEZ | Date: June 2026 | Stack: MongoDB · Express · React · Node.js

1. Solution Overview

ShopEEZ is a fully decoupled, full-stack MERN e-commerce web application providing:

- A premium dark-themed, responsive frontend built with React 19 + Vite + Redux Toolkit
- A RESTful backend API built with Node.js + Express.js (ES Modules)
- A MongoDB Atlas cloud database managed via Mongoose ODM
- JWT + Bcrypt for stateless authentication and role-based authorization
- A complete Admin Dashboard for real-time store management

2. Architecture Overview

React 19 (Vite) Frontend [Port 5173]

Redux Toolkit | React Router v6 | Axios JWT Interceptor | Vanilla CSS Dark Theme
!“ HTTP REST (JSON) via Axios

Node.js + Express.js Backend [Port 5000]

Routes: /api/auth /api/products /api/cart /api/orders /api/admin
Middleware: protect (JWT verify), admin (role check)
!“ Mongoose ODM

MongoDB Atlas Cluster

Collections: users, products, orders, carts, reviews

3. Key Design Decisions

Decision	Choice	Reason
Frontend build tool	Vite	Fast HMR, lightweight production builds
State management	Redux Toolkit	Predictable, scalable state for cart, auth, products, wishlist
CSS approach	Vanilla CSS	Full control over dark theme variables without framework bloat
Auth strategy	JWT Bearer tokens	Stateless auth ideal for decoupled SPA + API architecture
DB hosting	MongoDB Atlas	Managed cloud cluster with auto-scaling and built-in backups
Image handling	Multer + /uploads/	Local storage during dev; Cloudinary CDN in production

4. Frontend Route Map

Route	Page	Access
/	Home — Featured + Catalog	Public
/products	Product List + Filters	Public
/product/:id	Product Detail	Public
/login	Login	Public

/register	Register	Public
/wishlist	Wishlist	Public
/cart	Shopping Cart	Auth Required
/checkout	Checkout	Auth Required
/orders	Order History	Auth Required
/profile	User Profile	Auth Required
/admin/dashboard	Admin Dashboard	Admin Only
/admin/products	Manage Products	Admin Only
/admin/orders	Manage Orders	Admin Only
/admin/users	Manage Users	Admin Only